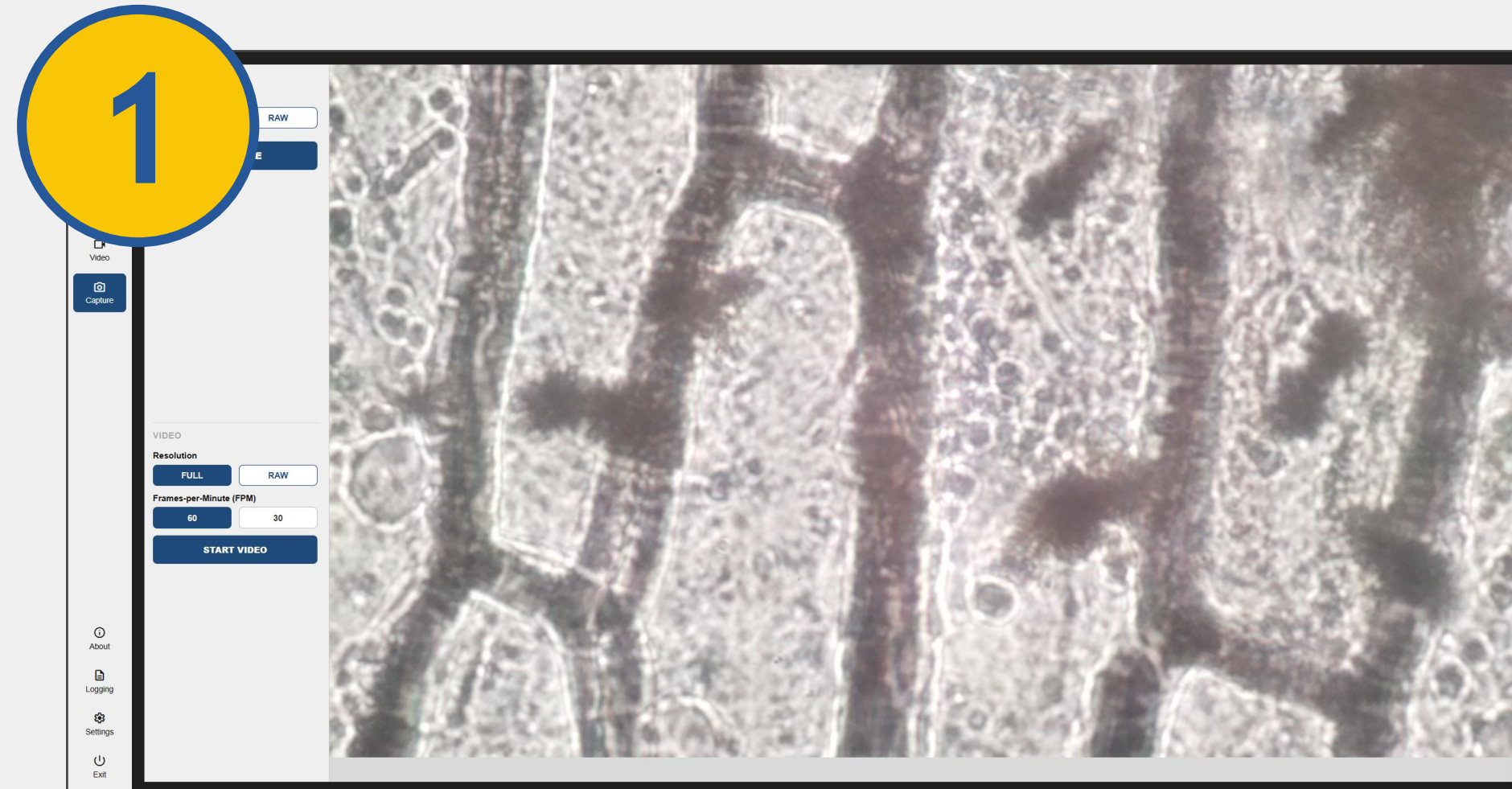
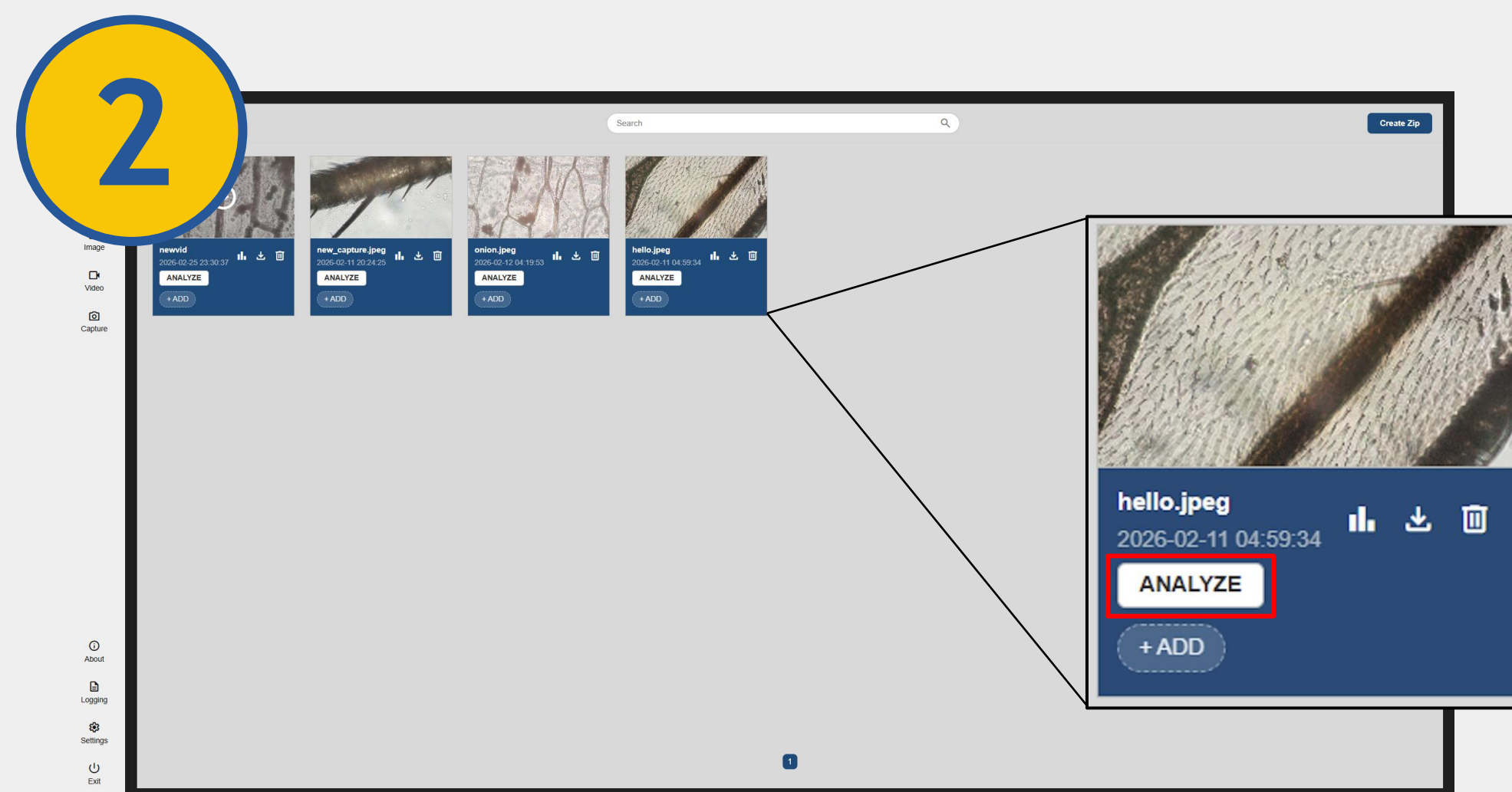




First, users capture an image or video from the microscope.

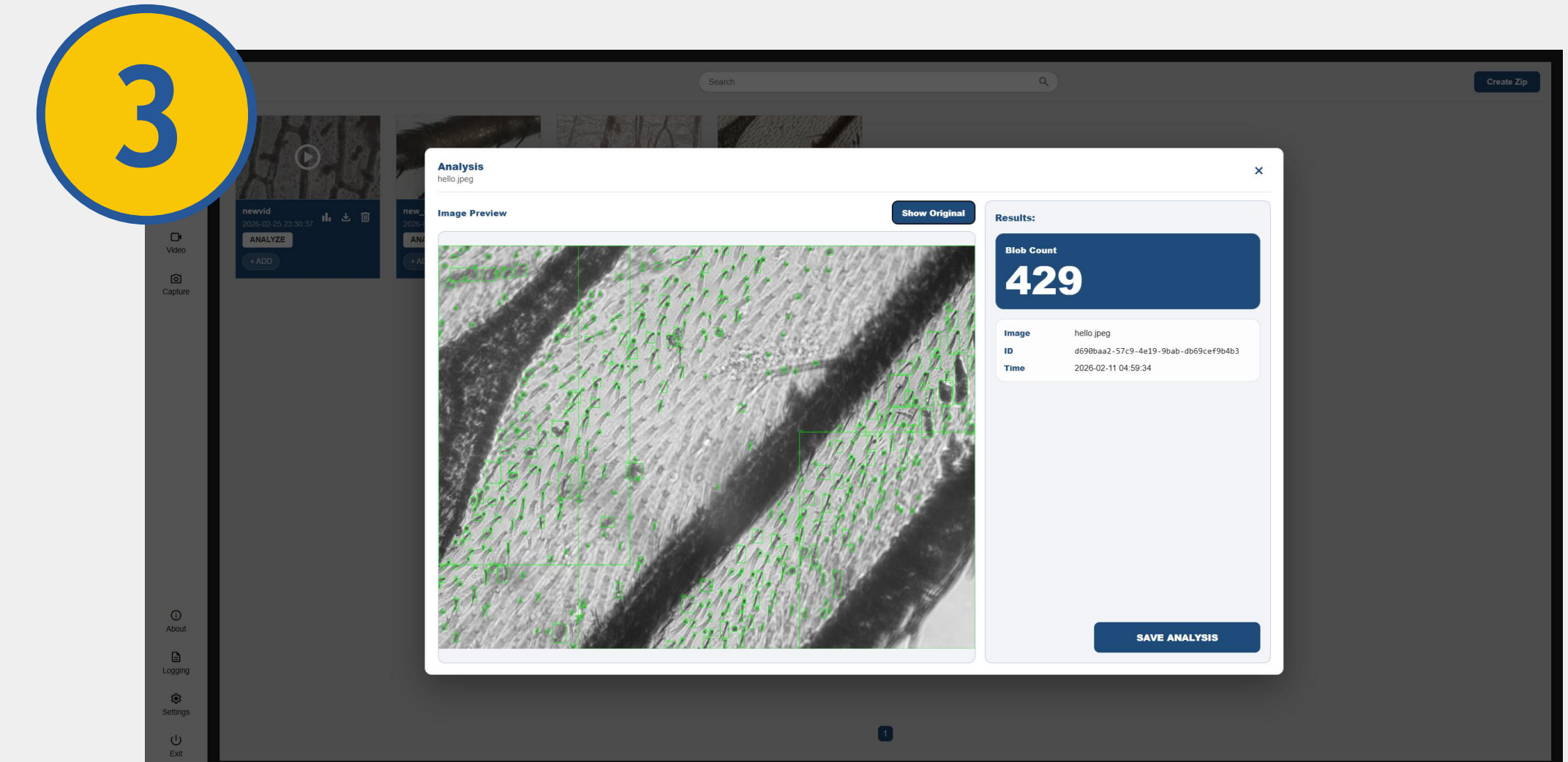


Next, users can select the Analyze feature to start our computer vision and machine learning analyses.

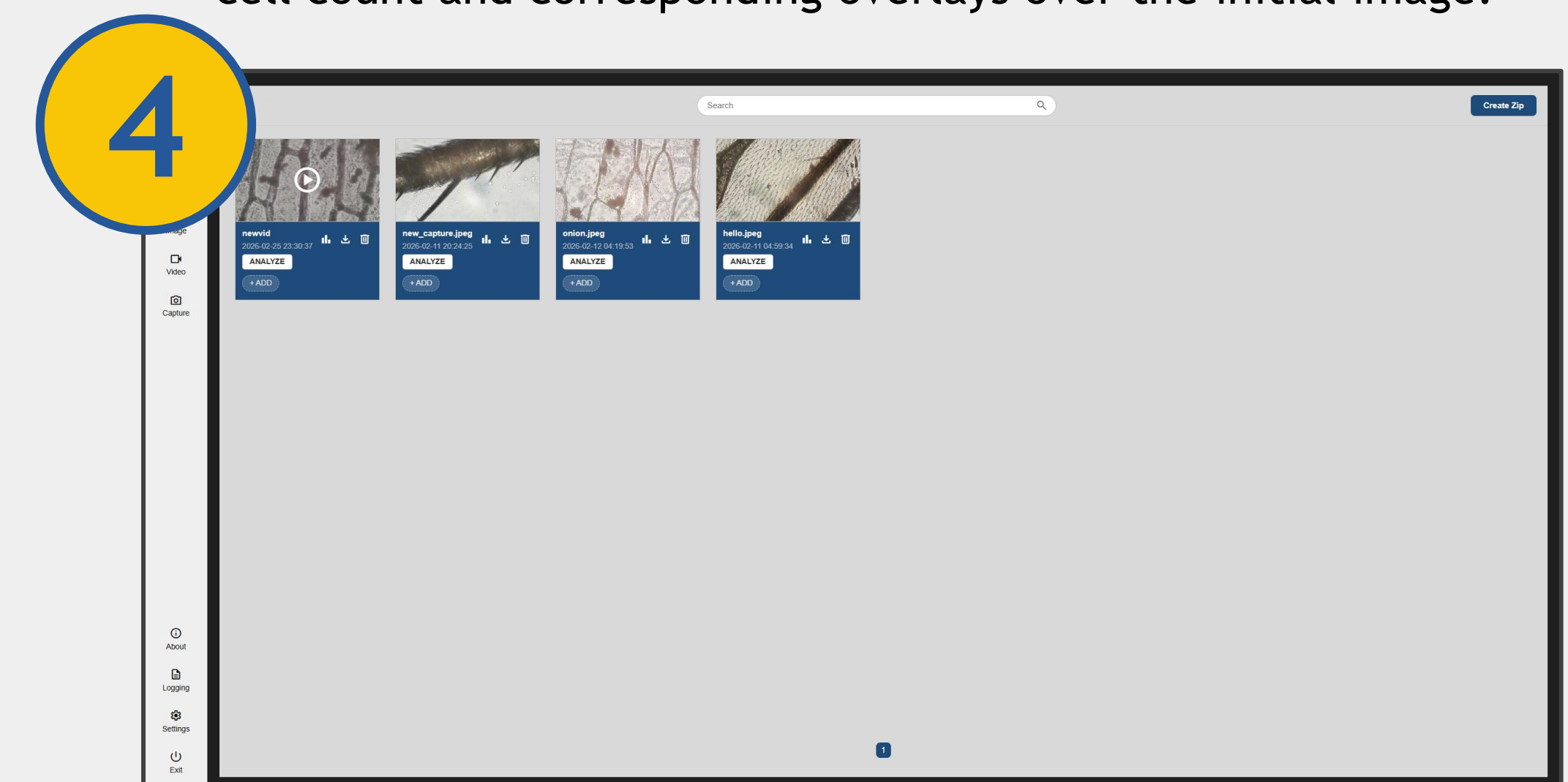


Our 3D printed microscope combined with our custom *Cellular Imaging Studio* software, based on *The OpenFlexure Project*, gives students, educators, and research professionals a low-cost solution for capturing high resolution cellular imagery, leveraging our AI to count cells and track their movement, and exporting the results.

Built with  openflexure
 project



Then, the user sees quantitative results, in the form of a total cell count and corresponding overlays over the initial image.



Captured images and videos and AI analysis results are saved in the Gallery for viewing and downloading - including saving raw analysis data in CSV format.

Objectives

- Develop a low-cost, accessible imaging system
- Enable real-time imaging, recording & stage control
- Ensure compatibility with *OpenFlexure Connect* interface
- Automate cell counting & motion tracking
- Create integrated software interface
- Support data & content export

Hardware

- Raspberry Pi 4 Model B
- Raspberry Pi Camera Board v2
- *Sangaboard* v5 controller for motors and LED
- 28BYJ-48 5V stepper motors (3)
- BXEN-50S-11M-3C LED board
- 40X plan objective for magnification
- Achromatic lens for improved color and clarity
- 3D-printed body and components

Software

- OpenCV: AI/ML image processing
- OpenFlexure API: Hardware programming
- Requests: Hardware to Server connectivity
- FastAPI: Server to Frontend connectivity
- Uvicorn: Web server for testing API functionality
- CSV: Reading and writing CSV files
- Vue: Microscope UI development
- Npm: node package manager for Javascript

Professional Standards

- GPIO, I2C, and SPI for hardware controls
- MIPI Alliance CSI-2 for camera connection
- IEEE 802.3 Standards for Ethernet connection
- RESTful API, HTTP, TCP, and JSON used for communications between client and servers
- ISO/IEC AI/ML standards for model type and implementation